

Price Change Dynamics across the Business Cycle: Evidence from Market Reference Prices*

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Abstract

This paper examines how frequently, and by how much, prices change across the U.S. business cycle. We analyse price rigidity across 1,110 retail product categories over 18 years using market reference prices, which we define as the modal price of identical products sold across all retail stores during the same week. We introduce a market reference price algorithm that more accurately measures aggregate nominal rigidity in the retail sector. The algorithm is implemented across the entire NielsenIQ retail scanner dataset, which contains granular data from over \$4.8 trillion worth of retail sales between 2006–2023. We find that: (1) aggregate market reference prices are only mildly sticky, but there is substantial heterogeneity in price stickiness across retail products; (2) while posted price-change frequency exhibits a strong upward trend over 2006–2023, market reference price-change frequency remains largely stable; and (3) in low-inflation periods, variations in inflation are primarily accounted for by changes in the size of price adjustments rather than their frequency, whereas in high-inflation periods, the frequency of price increases becomes the dominant margin of adjustment.

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I Introduction

The frequency and size of price changes are central to macroeconomic theory. If prices change often, nominal shocks such as monetary policy will quickly pass through the economy and impart minimal real economic effects. In contrast, if prices are highly rigid and instead change only slowly, these same nominal shocks will significantly impact the real economy. Macroeconomic theory shows that the distribution of price change sizes also shapes the real effects of nominal shocks that propagate through the economy. These price-change dynamics determine the effectiveness of monetary policy as a macroeconomic stabilisation tool during periods of economic contraction, such as the Great Recession. They also determine how significant the welfare costs will be of contractionary monetary policy actions taken to try to tame inflationary surges, such as the 2021–2023 high-inflation episode. In this paper, we examine price-change dynamics in the U.S. retail sector between 2006–2023, addressing this central question in macroeconomic theory.

The rapid digitalisation of the U.S. economy since the turn of the millennium has endowed consumers and retailers with real-time access to the nationwide prevailing price of retail products. We denote this concept as a market reference price and define it as the modal price of all identical products that are sold in all retail stores across the United States during the same week. In this paper, we use market reference prices to analyse the development of retail price-change dynamics across the full U.S. business cycle. We introduce a market reference price algorithm that enables the accurate calculation of price rigidity measurements for the U.S. retail sector. We demonstrate that market reference prices reveal the core structural dynamics of price changes which occur in response to changing monetary and economic conditions. This analysis enables us to identify several new stylised facts about U.S. price rigidity and their implications for New Keynesian macroeconomic theory.

This study of price-setting represents the largest analysis of price micro-data ever undertaken¹. This paper analyses the entire NielsenIQ retail scanner dataset, which contains data on all weekly transactions made at 71,249 retail stores throughout every major market in the United States. Altogether, after data cleaning and sparsity removal, this paper analyses over 239 billion price observations from over 2.26 million unique products which totalled over \$4.8 trillion worth of retail sales between 2006–2023. The enormous scale of the dataset analysed in this study comes with several powerful benefits. Firstly, it enables the calculation of accurate market reference prices as the dataset covers over 50% of all U.S. grocery and drugstore retail transactions and more than 30% of all U.S. mass-merchandise retail transactions. Secondly, price rigidity can be measured across a broad cross-section of the economy because over 1,100 categories of

¹This study analyses over 10.35 terabytes of price micro-data from the NielsenIQ retail scanner dataset, which was made possible through the use of a supercomputer with 128 central processing unit cores and 1 terabyte of random-access memory.

retail products are analysed. Thirdly, we can calculate highly precise aggregate price rigidity measurements that are reflective of the entire country, because they are based on more than 239 billion price observations collected from 71,249 stores located across all major U.S. retail markets. Lastly, we can analyse the development of price-change dynamics across the full U.S. business cycle, as we examine an 18-year period characterised by both economic contractions and expansions and episodes of low and high inflation. In Section II, we provide a detailed description of the NielsenIQ retail scanner dataset.

In Section III, we formally establish the concept of a market reference price and examine its role in shaping U.S. retailer price-setting behaviour. Across all retail products, we find that 35.6% of prices exactly match their relative market reference price, highlighting the strong gravitational pull that market reference prices exert. Furthermore, nearly half (46.2%) of all prices are within $\pm 5\%$ of their relative market reference price. We also find that market reference prices have a very strong tendency to rest at digit-9-ending price points (79.6%) and to systematically transition between them. These patterns indicate that across the retail sector, prices continuously orbit their relative market reference prices, adjusting in conjunction with them. In Section IV, we discuss why price rigidity measurements are highly sensitive to the choice of price-filtering algorithm and introduce a novel market reference price-filtering algorithm to the literature. The proposed market reference price algorithm provides a simple and effective mechanism to purge temporary price deviations from the data and therefore enable accurate measurements of price stickiness in the U.S. retail sector.

In Section V, we find that prices are only mildly sticky in the U.S. retail sector: the median weekly market reference price-change frequency across products is 16.38%. This implies that the retail sector is a smaller source of monetary non-neutrality relative to other sectors of the U.S. economy. This finding contrasts with that of Eichenbaum et al. (2011), who postulated that nominal rigidities in the retail sector take the form of significant inertia in their reference price measure. It also contrasts with the claim made in Kehoe and Midrigan (2015) that prices are sticky after all, based on their measurements of retail scanner data using their regular price measure. However, price-changes are not independent over time; a subset of products undergo repeated weekly price adjustments, while many others remain unchanged for prolonged periods: the median share of retail products experiencing at least one weekly market reference price-change within a month is 41.55%. This enhances the overall degree of price stickiness in the retail sector. Moreover, there is also substantial heterogeneity in the weekly frequency of market reference price-changes across retail products, ranging from 0.3% for magazines to 43.2% for fresh strawberries. Some retail product categories exhibit volatile price dynamics and contribute little to monetary non-neutrality, while others display much stickier price dynamics and contribute more significantly to the persistent real effects of monetary policy.

We also analyse the development of price rigidity in the U.S. retail sector over time. We find that there has been a strong upwards trend in the frequency of price-changes between 2006–2023. However, the frequency of market reference price-changes has remained largely stable over the same period. We attribute this finding to an increased prevalence in temporary price deviations likely driven by recent technological advancements. We also find that there was no apparent response in aggregate price rigidity during the Great Recession and no evidence of cyclicity in price stickiness.

In Section VI, we analyse price-change dynamics more granularly by decomposing retailer price-setting behaviour into four mechanical levers: the frequency of price increases, the frequency of price decreases, the average size of price increases and the average size of price decreases. We examine the dynamics of these four mechanical levers of inflation over the sample period. We find that in low-inflation periods, variations in inflation are primarily accounted for by changes in the size of price adjustments rather than their frequency; while in high-inflation periods, the frequency of price increases becomes the dominant margin of adjustment. We then discuss the implications of this finding for the microeconomic foundations of price rigidity underlying New Keynesian macroeconomic theory.

The development over time of increasingly large price micro-datasets has steadily improved our understanding of price rigidities. The first academic study which sought to measure the price-change frequency of goods in the United States using a price micro-dataset was [Cecchetti \(1986\)](#). In this paper, they collected data from various libraries on the newsstand prices of thirty-eight magazines from 1953–1979 and found that prices typically remained unchanged for periods ranging from 1.8 to 14 years. Another noteworthy early academic study in this literature was [Kashyap \(1995\)](#), who collected data on the prices of twelve retail goods between 1953–1987 by directly copying them out of old retail catalogues, and reported that prices are typically fixed for more than one year.

The first academic study to utilise a large price micro-dataset that is representative of the entire United States was [Bils and Klenow \(2004\)](#). In this paper, they analysed unpublished data from the Bureau of Labour Statistics (BLS) Commodities and Services Substitution Rate Table, which contains average price change frequencies and substitutions by disaggregated product categories between 1995–1997. This paper shattered the conventional wisdom by reporting that prices typically remained fixed for only 4.3 months. A significantly larger and more granular BLS price micro-dataset called the CPI Research Database (CPI-RDB) was used in [Nakamura and Steinsson \(2008\)](#) and [Klenow and Kryvtsov \(2008\)](#), which were milestone contributions to this literature. This BLS micro-dataset underlies the construction of the U.S. CPI inflation rate and is collected from monthly BLS surveys of the prices of around 85,000 goods from around the United States. [Klenow and Kryvtsov \(2008\)](#) used data from the CPI-RDB to establish eight

stylised facts², which they used to assess the robustness of the [Taylor \(1980\)](#), [Calvo \(1983\)](#), [Dotsey et al. \(1999\)](#) and [Golosov and Lucas \(2007\)](#) models. [Nakamura and Steinsson \(2008\)](#) also used CPI-RDB data from 1988–2005, totalling over 9 million observations, to establish five empirical facts about price adjustment in the United States³.

The CPI-RDB was later extended back to 1977 by scanning and digitising physical microfilm cartridges that had been archived at the former Washington D.C. BLS headquarters. [Nakamura et al. \(2018\)](#) used this extended micro-dataset to study price-setting behaviour and dispersion during the Great Inflation. Previous major studies of aggregate U.S. price stickiness had solely focused on periods characterised by low inflation and reduced macroeconomic volatility, such as the Great Moderation. Using the extended CPI-RDB, they showed that the mean absolute size of price changes remained flat between 1978–2014. Consequently, they found no evidence that price dispersion rose during the Great Inflation and concluded that the standard New Keynesian analysis of the welfare costs of inflation is wrong. In this paper, we document similar price-change dynamics in the U.S. retail sector during the high inflation episode between 2021–2023.

Surveys of this empirical literature and the stylised facts about sticky prices that it has documented, can be found in [Klenow and Malin \(2010\)](#) and [Nakamura and Steinsson \(2013\)](#). This work to establish accurate stylised facts about sticky prices has helped guide the development of macroeconomic models. New Keynesian Dynamic Stochastic General Equilibrium (DSGE) models which seek to study the real effects of structural economic shocks, must be underpinned by robust and empirically disciplined microeconomic foundations of price-setting. Structural economic shocks are characterised by their interactions with frictions that buffer and shape their propagation throughout the economy. When considering the real economic effects of monetary policy, the most important of these frictions are those associated with the adjustment of prices. The presence of these frictions induces rigidity in nominal prices, which is known as price stickiness. Therefore, the frequency and size of price changes are central issues in New Keynesian macroeconomic theory.

New Keynesian macroeconomic theory emerged during the 1980s amid a flurry of proposed theoretical frameworks explaining how nominal prices could be sticky. These frameworks can be classified into two different styles of models: time-dependent sticky price models and state-dependent sticky price models. In time-dependent sticky price models such as [Calvo \(1983\)](#), a fixed fraction of firms adjusts their prices each period independent of economic

²These eight facts are: (1) Frequent price changes, (2) Large average absolute price changes, (3) Many small price changes, (4) Variable price durations, (5) Flat hazard rates, (6) Size of price changes does not increase with duration, (7) Intensive margin dominates the variance of inflation, (8) Frequency of price increases and decreases moves with inflation.

³These five facts are: (1) For consumer prices, the median frequency of nonsale price change is roughly half of what it is including sales. (2) One-third of nonsale price changes are price decreases. (3) The frequency of price increases covaries strongly with inflation. (4) The frequency of price-change is highly seasonal. (5) They find no evidence of upward-sloping hazard functions of price-changes for individual products.

conditions. The key advantage of the [Calvo \(1983\)](#) framework is its tractability and for this reason it remains the workhorse framework of contemporary New Keynesian DSGE models. In contrast, state-dependent sticky price models assume that all firms can adjust their prices each period subject to payment of adjustment costs and the state of the economy. Therefore, the firms that adjust their prices will be the ones for whom it will be most profitable. Prominent examples include the [Rotemberg \(1982\)](#) model, where firms incur quadratic costs when adjusting prices, and the [Golosov and Lucas \(2007\)](#) menu cost model, which instead assumes fixed adjustment costs.

In this paper, we analyse price-change dynamics across the full U.S. business cycle to assess which models best capture actual U.S. price-setting behaviour. We determine that while the [Calvo \(1983\)](#) time-dependent price-setting framework can adequately approximate the observed price-setting behaviour during low-inflation periods, it struggles to capture our finding that in high-inflation periods, the frequency of price increases becomes the dominant margin of adjustment. In contrast, state-dependent menu cost models can better reflect that when inflation is high, more retailers will increase their prices because the benefit derived from adjusting to the new optimal price will exceed the adjustment cost. These findings imply that monetary policy rules derived under the [Calvo \(1983\)](#) price-setting framework may overstate the welfare gains from aggressive price stabilisation during high-inflation episodes.

The paper is organised as follows. Section [II](#) describes the NielsenIQ retail scanner dataset. Section [III](#) establishes the concept of a market reference price. Section [IV](#) outlines the reasons why the proposed market reference price algorithm outperforms the price-filtering algorithms previously employed in the prior literature. Section [V](#) measures price rigidity across product categories and time. Section [VI](#) analyses price-change dynamics across the business cycle. Section [VII](#) concludes.

II Data

In the United States, product manufacturers obtain barcodes called Universal Product Codes (UPCs) from GS1⁴ which uniquely identify their product. The manufacturer then distributes their product to retailers with the UPCs attached. When the product is scanned at a point-of-sale system, such as an electronic cash register, data about the transaction is stored in scanner datasets. The most significant scanner dataset that is publicly available to researchers is the NielsenIQ Retail Scanner dataset. It contains weekly pricing, sales volume and store environment data from over 71,000 stores operated by more than 90 participating retail chains. These retail chains include grocery, drug, mass merchandise, convenience and liquor stores which cover the entire

⁴GS1 is a non-profit global standards body that administers product identification and barcode systems across more than 150 countries.

United States. Altogether, the NielsenIQ Retail Scanner dataset contains more than 50% of the total sales volume of all U.S. grocery and drug stores and more than 30% of the total sales volume of all U.S. mass merchandise stores. Weekly product data for over 2.26 million UPCs are collected which are organised into over 1,100 product modules⁵, which are categorised into 125 product groups and 10 product departments. These 10 product departments are Health and Beauty Aids, Dry Grocery, Frozen Foods, Dairy, Deli, Packaged Meat, Fresh Produce, Non-Food Grocery, Alcohol and General Merchandise. After data cleaning and sparsity removal, this paper analyses 239,648,464,792 price observations, dwarfing all the price micro-datasets that have been previously used to measure U.S. price rigidities.

There are two alternative sources of price micro-data which have been frequently used in the literature to measure U.S. price rigidities: web-scraped prices from online retailers⁶ such as the dataset used in [Cavallo \(2018\)](#) and the micro-data underlying the Consumer Price Index⁷ (CPI) compiled by the Bureau of Labor Statistics (BLS) which was used in [Bils and Klenow \(2004\)](#); [Klenow and Kryvtsov \(2008\)](#); [Nakamura and Steinsson \(2008\)](#).

The application of web-scraping software to collect prices from online retailer’s webpages was pioneered by the Billion Prices Project academic research initiative, which is described in [Cavallo and Rigobon \(2016\)](#). The key advantage of the NielsenIQ retail scanner dataset over web-scraped price micro-datasets is its substantially broader coverage of product categories. As a result, it is more suitable for analysing price-setting behaviour across the entire U.S. retail economy. Additionally, the NielsenIQ retail scanner dataset spans a significantly longer time period than any publicly available web-scraped price micro-dataset. Consequently, it provides a more reliable basis for analysing price-setting behaviour over the full U.S. business cycle. Moreover, because the NielsenIQ retail scanner dataset is collected from physical stores, where the vast majority of U.S. retail sales still occur, it provides more accurate estimates of retail price-setting behaviour. The one clear disadvantage of the NielsenIQ retail scanner dataset relative to web-scraped price microdata is its reliance on weekly average prices. However, any potential bias arising from this practice is effectively mitigated by the proposed market reference price algorithm, which is discussed in Section IV.

Many important contributions to the U.S. sticky price literature are based on BLS price micro-datasets, such as the CPI-RDB. The key advantage of the NielsenIQ retail scanner dataset over the BLS CPI price micro-data is that it is sampled at a weekly frequency rather than a monthly

⁵Two examples of product modules are the Video and Computer Games product module in the General Merchandise product department and the Fresh Strawberries product module in the Fresh Produce product department.

⁶The scraping process involves daily scanning of the code of publicly available websites, identifying relevant pieces of information and storing them in a database.

⁷To construct the non-shelter portion of the CPI, the BLS surveys the prices of about 85,000 items a month. Individual prices are collected by around 400 BLS employees visiting 20,000 retail outlets a month, mainly across 45 large urban areas. They price each item for up to five years, after which the item is rotated out of the sample.

frequency⁸. [Cavallo \(2018\)](#) showed that monthly sampling tends to upwardly bias price stickiness measurements because many price-changes within the month are not observed. In contrast, they found that daily and weekly sampled data produce similar price stickiness measurements because few products have more than one price-change per week. Moreover, [Alvarez et al. \(2025\)](#) show that price-change hazard function spikes occur systematically every seven days which suggests that firms typically make many pricing decisions at weekly intervals. They also concluded that weekly price observations capture the core patterns in price-setting behaviour. Another important advantage of the NielsenIQ retail scanner dataset is that it does not suffer from any bias associated with the imputation of missing prices. The BLS commonly imputes prices into its micro-datasets due to product substitutions or temporarily missing prices with cell-relative imputation, a method that uses the average price-change within related categories of goods. However, there is one important disadvantage of the NielsenIQ retail scanner dataset in that it does not cover all expenditure categories of the CPI. In particular, the NielsenIQ Retail Scanner dataset does not contain prices for services, whose relative importance (excluding housing costs) made up (28.88%) of the CPI between 2006–2023⁹. Moreover, [Nakamura and Steinsson \(2008\)](#) reported that the services (excluding travel) component of the CPI has a lower price-change frequency and prevalence of sales than the aggregate CPI¹⁰.

III Establishing Market Reference Prices

This section seeks to examine the role that market reference prices play in shaping U.S. retailer price-setting behaviour. Market reference prices are defined as the modal price of all identical products that are sold in all retail stores across the United States during the same week. With the rapid digitalisation of the U.S. economy, consumers and retailers have become more informed about the prices at which products are most commonly sold nationwide. Consequently, their purchasing and price-setting behaviour have become increasingly anchored to the prevailing nationwide market reference price.

[DellaVigna and Gentzkow \(2019\)](#) show that most U.S. retail chains post nearly uniform prices for a retail product across all of their retail stores, irrespective of variation in local economic

⁸The BLS collects consumer prices monthly for food and fuel items in all areas. The BLS also collects prices monthly for all items in the three most populous areas: New York, Los Angeles and Chicago. The BLS collects prices for items in other categories and other urban areas only bimonthly.

⁹This figure is based on the author's calculations from the BLS relative importance tables for the CPI between 2006–2023. The figure is calculated as the mean importance weight of the services less rent of shelter component in the special aggregate indexes. The equivalent figure for total services (including housing costs) is (61.5%).

¹⁰Nakamura and Steinsson (2008) reported that for the period between 1998–2005, the expenditure weighted mean price-change frequency of filtered non-sale prices and unfiltered prices was (8.8%) and (9.1%) respectively for the services (excluding travel) CPI component. The equivalent figures for all sectors were (21.1%) and (26.5%) respectively. The expenditure weighted mean fraction of price-changes that are due to sales was (3.1%) for services (excluding travel) and (21.5%) for all sectors.

conditions and competitive pressures. Consequently, a retail product’s market reference price is determined by the price-setting behaviour of a small set of large retail chains. This intuitively clear concept, of a prevailing nationwide market reference price for a retail product, is rooted in the theory of coordination failures in price-setting amongst retailers. [Blinder et al. \(1998\)](#) found in their survey of two hundred U.S. firms, that out of twelve proposed theories for why prices are sticky, coordination failure was the one that received the most support¹¹. The reason that was predominantly provided for why coordination failure is an important source of price stickiness was that: “If we raised our prices first, we would lose too much business to low-price competitors”. In a competitive retail environment, such pressures can delay price increases, inducing prices nationwide to cluster at the market reference price until a price leader adjusts, thus creating stickiness in prices.

Another important factor contributing to nationwide price clustering at the market reference price is manufacturers setting recommended retail prices (RRPs) for their products. A clear example of this can be seen in [Figure I](#), which presents the nationwide posted prices and market reference price of the most popular video game in the dataset, which was sold across 2,485 retail stores. The posted prices are the actual prices recorded that week at the electronic point-of-sale systems at every retail store where the video game was purchased¹². The market reference price is then calculated as the mode of all the posted prices recorded for the video game that week.

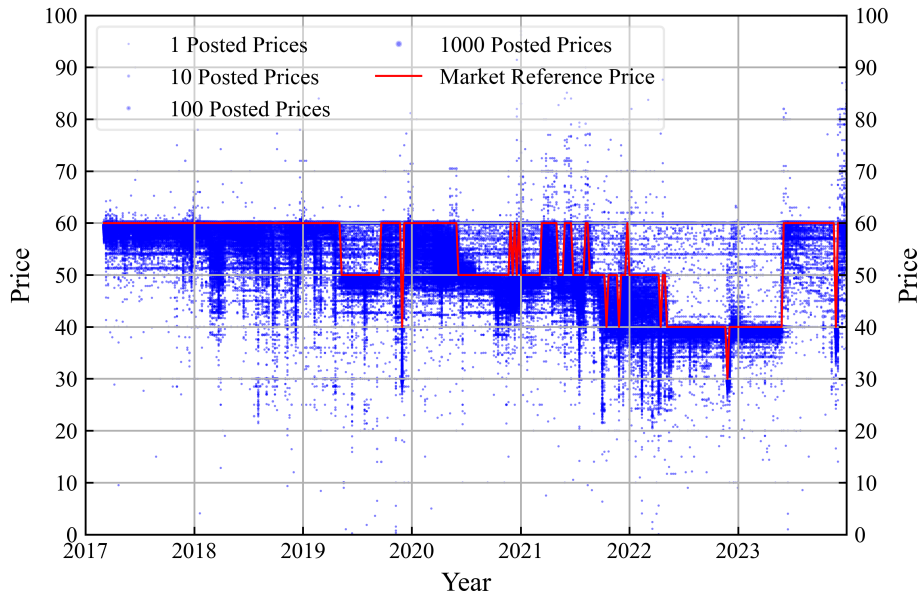
Across 198 of the 357 weeks in which the video game recorded transactions, its market reference price was \$59.99, clearly aligning with the RRP set by the video game publisher. In a further 98 weeks, the market reference price declined to \$49.99, likely reflecting strategic pricing intended to stimulate demand during the summer months, when sales are seasonally weaker, or to clear inventory and shelf space ahead of the release of a new, closely related title. In a low inflationary environment, RRP’s may exhibit inertia, causing posted prices to become firmly stuck to their relative market reference price for extended periods. As a result, regular consumers of the product become highly informed of the prevailing market reference price. When this occurs, clearly marked temporary price reductions can generate substantial surges in sales volumes, as consumers readily recognize the value of the discount¹³. Consistent with this mechanism, when the video game’s market reference price dropped to \$29.99 in the week ending 26 November 2022, which coincided with Black Friday, weekly sales across all stores increased to 45,664 units, approximately 10.1 times the pre-reduction weekly average of 4,513 units. Conversely, this same mechanism can also make retailers reticent to unilaterally increase their prices, due to fears of losing substantial market share to their competitors.

¹¹In fact, 71.4% of the retail firms surveyed reported that coordination failures were either moderately important (14.3%) or very important (57.1%) in explaining price stickiness.

¹²This definition follows the standard convention in the literature, established since [Klenow and Kryvtsov \(2008\)](#), where posted prices are used to denote the raw, actual prices to clearly distinguish them from filtered prices.

¹³[Abe and Tonogi \(2010\)](#) provided a notable example of this mechanism: when a Japanese retail store reduced the price of a cup noodle product from 140 yen to 84 yen in September 2005, daily sales increased twenty-five-fold relative to the pre-discount average.

Figure I: Nationwide Prices of a Video Game



Notes: This figure presents a scatter plot of all store-level weekly posted prices (blue dots) alongside the calculated weekly market reference price (red line) of the product UPC with the most observations within the Video and Computer Games product module in the NielsenIQ retail scanner dataset. Marker size is scaled so that price points observed at more stores in a given week appear as larger dots.

The repeated transitions in the video game's market reference price between digit-9-ending price points are indicative of a deliberate pricing strategy¹⁴. These digit-9-endings are price points which firms strategically adopt to leverage consumers' cognitive biases¹⁵. This pricing strategy, known as charm pricing, subtly influences consumers' purchasing behaviour by exploiting their tendency to read prices from left to right¹⁶. In general, the market reference prices of retail products in the United States are heavily concentrated at these price points. Figure II presents the frequency distribution of the last decimal digit of posted prices and market reference prices of all retail products in the NielsenIQ retail scanner dataset. The concentration at price points is striking: 56.9% of posted prices and 79.6% of market reference prices are set at digit-9-ending price points. A key rationale for this large difference in price point concentrations is that retailers regularly conduct temporary price discounts below the market reference price before returning exactly to that same market reference price¹⁷.

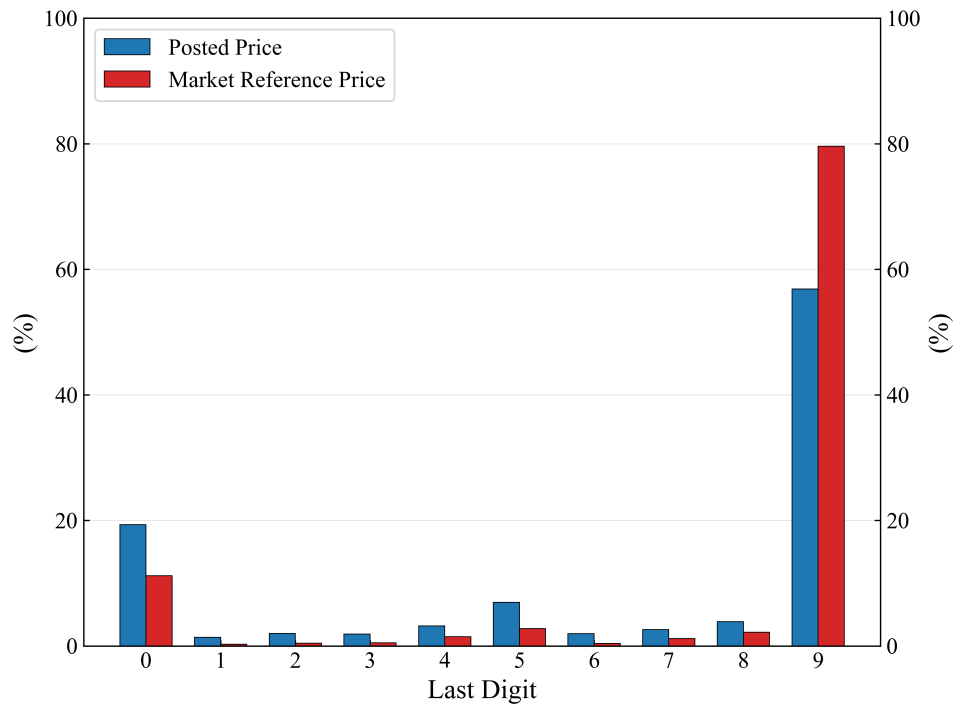
¹⁴Blinder et al.'s (1998) price stickiness survey of U.S. firms found strong support for the price point theory of price stickiness among retailers who sell products directly to consumers.

¹⁵In general, price points rest at kinks in consumer demand curves and thus also kinks in the firm's profit functions which endows them with the property of being local profit maxima.

¹⁶Strulov-Shlain (2023) estimated the magnitude of this left-digit bias and found that consumers respond to a one-cent increase from a 99-ending price as if it were more than a twenty-cent increase.

¹⁷Knotek II (2024) found that post-sales prices are more likely to return to their pre-sale price level if that price was a digit-9-ending price point.

Figure II: Last Digit Distribution of Prices



Notes: This figure presents a grouped bar chart showing the percentage distribution of the last digit price endings for both posted prices (blue bars) and market reference prices (red bars) across all observations in the NielsenIQ retail scanner dataset.

The role of digit-9-ending price points in manifesting price rigidities was studied by [Levy et al. \(2011\)](#), who found that digit-9-ending prices are less likely to change than non-digit-9-ending prices and that the average size of price changes is larger for digit-9-ending prices than non-digit-9-ending prices. A key reason for these patterns is the strong tendency of market reference prices to transition between digit-9-ending price points. In the NielsenIQ retail scanner dataset, 56.9% of all market reference price changes occur between digit-9-ending prices. Additionally, 78.1% of all market reference price changes away from a digit-9-ending price are to another digit-9-ending price. Therefore, we can clearly see that market reference price changes are not random; retailers systematically adjust their prices between digit-9-ending price points to maximise their profits.

For relatively expensive, low-turnover retail products such as consumer electronics or durables, modern U.S. consumers will routinely research nationwide prices online before purchasing. As a result, price dispersion for these products is compressed, even in markets where favourable local economic conditions and limited competitive pressures would otherwise allow retailers to sustain larger retail margins. This widespread price transparency induces posted prices to gravitate towards their relative market reference price. However, for cheaper, high-turnover categories of products, the expected gains from price searching are limited, weakening the gravitational pull

of their relative market reference prices. A natural metric of this gravitational pull is the share of posted prices that exactly match their relative market reference price.

Table I reports, by product department and for the aggregate total, the shares of posted prices equal to, below, or above the market reference price, along with the corresponding shares of revenue generated at those price levels, in the NielsenIQ retail scanner dataset. Overall, 35.6% of posted prices exactly match their relative market reference price, highlighting the strong gravitational pull that market reference prices exert. Across product departments, the strength of this gravitational pull varies, with the share of posted prices exactly matching the market reference price ranging from 27.5% for Alcohol to 57.5% for General Merchandise. Even in perishable product departments such as Fresh Produce, Packaged Meat and Dairy, where short product lifespans lead to frequent discounting and substantial revenue shares being generated below the market reference price, the high rates of exact price matching indicate that market reference prices nonetheless provide a valuable central measure of nationwide prevailing prices.

Table I: Price Matching and Revenue Share by Product Department

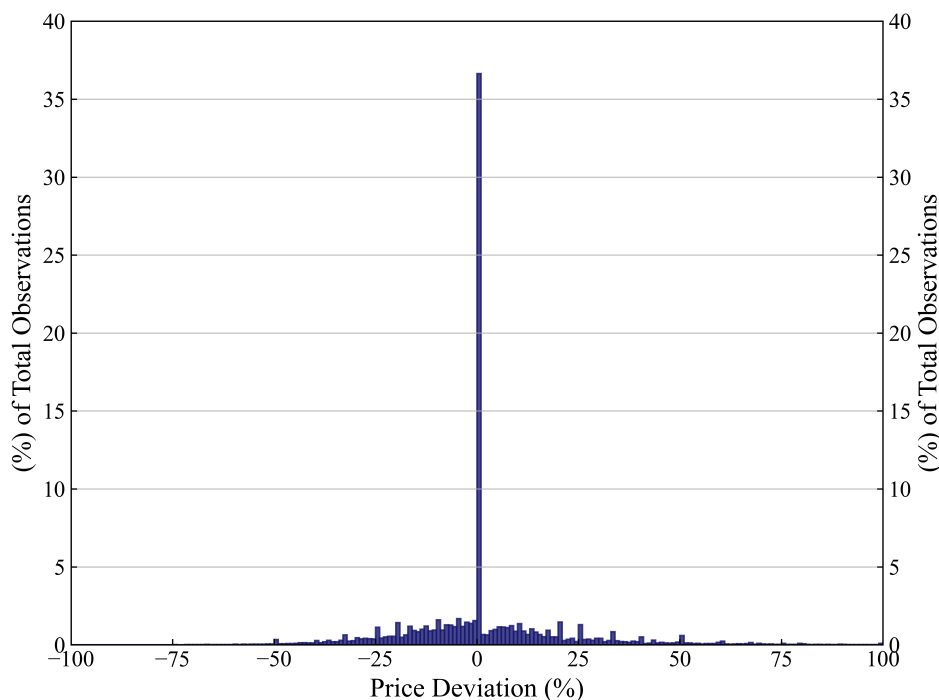
Department	Price Match (%)			Revenue Share (%)			Total (M)
	At MRP	Below	Above	At MRP	Below	Above	Total (M)
Health and Beauty Aids	36.8	34.8	28.4	32.9	39.6	27.5	44372.3
Dry Grocery	33.1	34.0	32.9	26.4	43.5	30.2	110708.9
Frozen Foods	31.9	34.8	33.3	29.2	40.7	30.1	16630.5
Dairy	30.8	36.1	33.1	23.0	45.9	31.1	10808.8
Deli	37.8	35.0	27.2	34.4	41.5	24.0	3604.1
Packaged Meat	35.1	35.8	29.1	27.6	46.1	26.3	3668.0
Fresh Produce	43.3	34.0	22.7	31.6	44.2	24.3	1849.7
Non-Food Grocery	38.5	30.0	31.5	27.1	39.8	33.1	27291.5
Alcohol	27.5	35.7	36.9	22.7	40.0	37.3	6917.7
General Merchandise	57.5	23.9	18.6	52.9	31.4	15.7	13796.9
TOTAL	35.6	33.3	31.0	28.6	41.8	29.6	239648.5

Notes: Price Match columns show the percentage of observations where the posted price equals (At MRP), is below, or is above the market reference price. Revenue Share columns show the percentage of total revenue generated at each price point. Total Observations are in millions. Data covers 2006–2023. January 2006 excluded.

In general, the posted prices of retail products will cluster around their relative market reference price. Figure III presents the distribution of percentage deviations of all posted prices away from their relative market reference price, in the NielsenIQ retail scanner dataset. Nearly half (46.28%) of all posted prices are within $\pm 5\%$ of the market reference price. 57.68% of posted prices are within $\pm 10\%$ of the market reference price and over three quarters (75.56%) are within $\pm 20\%$ of the market reference price. Across product departments, the share of posted prices within $\pm 5\%$ of the market reference price ranges from 37.38% in the Alcohol product department to 66.30% in the General Merchandise product department. Overall, these patterns indicate that across the retail sector, posted prices continuously orbit their relative market reference prices, adjusting in

conjunction with them. Consequently, we can conclude that market reference prices reveal the core structural dynamics of price-changes which occur in response to changing monetary and economic conditions. This conclusion enables the design of a market reference price algorithm that can accurately measure aggregate price stickiness, which we introduce in Section IV.

Figure III: Posted Price Deviation from their Market Reference Price



Notes: This figure presents a histogram of the percentage deviation of all the posted prices from their relative market reference price, for every UPC observation in the NielsenIQ retail scanner dataset over the period 2006–2023.

IV Price Filtering Algorithms

Measurements of price rigidity are highly sensitive to the choice of price-filtering algorithm, and consequently so are the inferred real economic effects of monetary policy. Therefore, before we seek to measure aggregate price rigidity in the U.S. retail sector across the business cycle, we must first establish the price-filtering algorithm that is most appropriate. This section outlines the reasons why the proposed market reference price algorithm outperforms the price-filtering algorithms previously employed in the literature. The seminal paper in this literature, [Bils and Klenow \(2004\)](#) considered all temporary sales as price-changes. As a result, they found that the median implied duration of prices was 4.3 months within the nonshelter portion of the CPI between 1995–1997. However, [Guimarães and Sheedy \(2011\)](#) showed that the real effects of monetary policy in a DSGE model with sticky prices and fully flexible sales are similar to

those found in a standard DSGE model with sticky prices and no sales¹⁸. They conclude from these estimates that sales are essentially irrelevant for monetary policy analysis. Nakamura and Steinsson (2008) found that after discarding temporary sales using the BLS sales flag the median implied duration of prices was 8.6 months within the same dataset used by Bils and Klenow (2004). Therefore, there is a clear need to filter these temporary sales out of the price rigidity measurements to obtain accurate results.

It has been well established in the economics, industrial organisation and marketing literatures that many price changes are temporary sales, which are characterised by large temporary decreases in the price of a good followed by a return to the exact same previous price. A multitude of rationales have been proposed to explain this price-setting behaviour including the following: temporary sales are used for intertemporal price discrimination in markets where consumers exhibit heterogeneous demand elasticities (Sobel 1984; Guimarães and Sheedy 2011), which may depend on whether they purchase for immediate consumption or for personal inventory stockpiling (Narasimhan et al. 1996; Ariga et al. 2001); they are strategically timed during periods of high footfall such as holidays or weekends to maximise sales volumes (Warner and Barsky 1995) and to implement loss-leader pricing strategies (Chevalier et al. 2003); they are employed to clear surplus store inventory (Lazear 1986; Hosken and Reiffen 2004); and they incur lower menu costs than permanent price changes (Kehoe and Midrigan 2015).

Temporary price increases are also found in price microdata, albeit at a less frequent rate¹⁹. Rationale which can explain this pricing behaviour includes the following: dynamic pricing practices²⁰; strategic signalling of high product quality for a newly introduced product; market testing, such as A/B testing, to estimate consumer demand for a product at a higher price before committing to a permanent change; managing inventory to prevent stock-outs by slowing demand until a new shipment of products is received²¹, and the pass-through of temporary increases in production costs to consumers in order to preserve retail margins²².

¹⁸The cumulated response of output to a monetary policy shock in their model with fully flexible sales is 89% of the cumulated response in the standard model. When the model is recalibrated to account for heterogeneity in the frequency of sales across sectors, this figure rises to 96%.

¹⁹When using the Dominick's dataset, Eichenbaum et al. (2011) and Chahrour (2011) found that when employing their respective price algorithm procedures, 21% and 12% of prices are above the corresponding reference price.

²⁰A salient example of this pricing practice is found in seasonal goods such as toys or gaming consoles before Christmas which have unpredictable consumer demand and long production times. A detailed discussion of this topic can be found in Çömez-Dolgan et al. (2020).

²¹Temporary delays in manufacturers delivering products to retailers can arise from temporary factory closures due to strikes or disruptions in the transfer of freight by shipping or delivery trucks due to adverse weather or geopolitical events.

²²Temporary increases in production costs can arise due to adverse weather or geopolitical events which impact the supplies of industrial metal, energy, agricultural commodities and intermediate goods such as semiconductor chips used in manufacturing processes or impede the supply of feed grain for livestock.

Nakamura and Steinsson (2008) introduced two slightly different V-shaped temporary sales filtering algorithms to the literature²³, which remove temporary sales from price rigidity calculations and have been widely adopted in other studies. For example, these algorithms have been previously implemented on a small scanner dataset called the Dominick's dataset²⁴ in Nakamura and Steinsson (2011), to estimate weekly price-change frequencies. Readers are referred to the supplementary materials of Nakamura and Steinsson (2008) for a comprehensive explanation of how these algorithms work. In general, V-shaped temporary sales filter algorithms, such as the one used in Nakamura and Steinsson (2008), identify temporary sales by looking for prices that initially fall and then return to the same previous or higher price level. However, these algorithms have an important limitation: they do not remove upward temporary deviations in prices, which results in an underestimation of price stickiness as discussed in Chahrour (2011) and Kehoe and Midrigan (2015).

Eichenbaum et al. (2011) alternatively propose that these nominal price rigidities instead take the form of inertia in reference prices, which are defined as the modal price of a product within a calendar quarter. They apply their reference price algorithm to scanner data from a large U.S. food and drug retailer and find that reference prices are particularly inertial, with an average duration of roughly one year, even though prices change roughly once every two weeks. These results imply that there is considerable nominal rigidity in the U.S. retail sector and that it is a major source of monetary non-neutrality. However, Chahrour (2011) reports that because the Eichenbaum et al. (2011) reference price algorithm restricts reference prices to change only on arbitrary calendar dates, it may therefore identify spurious sales and price spikes²⁵. Additionally, the algorithm considers only UPC-store pairs which appear in all quarters in the dataset and for at least 12 weeks in each of these quarters. This methodological approach results in the deletion of all data collected from 757 of the 1000 stores in their scanner dataset. These incredibly restrictive constraints would have removed most of the products from the NielsenIQ retail scanner dataset, including all seasonal goods. This approach would have created substantial sample-selection bias, especially given the significant degree of observed cross-sectional heterogeneity of price rigidities across products. The algorithm's heavy contortions of the data, along with the unrealistic estimates of price stickiness it generates, rendered it unsuitable for consideration.

The regular price algorithm proposed by Kehoe and Midrigan (2015) is somewhat conceptually similar to the reference price algorithm of Eichenbaum et al. (2011), as both involve the

²³Sales filter B is designed to remove only price patterns in which the price returns to the same price within n months or less. Sales filter A is designed to remove the price patterns removed by sales filter B as well as price patterns where a sale is followed by a change in the regular price.

²⁴The Dominick's dataset covers store-level scanner data collected at Dominick's Finer Foods which was a Chicago-area grocery store chain. Altogether, the dataset contains over 98 million weekly price observations from 29 different product categories throughout all stores in this 100-store chain. Store-level data on the sales of more than 3,500 UPCs were collected from 1989 to 1997.

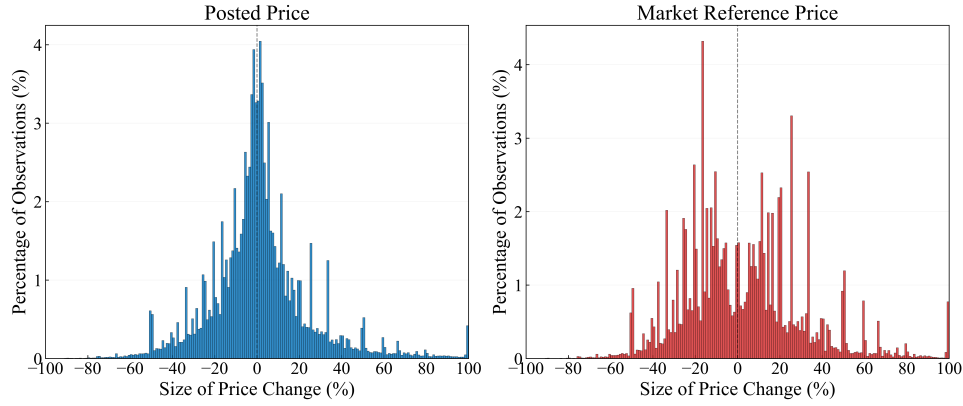
²⁵Chahrour (2011) also makes the argument that even if reference price decisions are made at a quarterly frequency, it seems unlikely that they will line up with any particular calendar definitions of these quarters.

calculation of a modal price. Readers are referred to the supplementary materials of [Midrigan \(2011\)](#) and [Kehoe and Midrigan \(2015\)](#) for comprehensive explanations of how this algorithm works. In general, the algorithm performs well for products with sticky prices such as cigarettes but struggles considerably to accurately capture the price dynamics of many products with more volatile prices such as fresh berries or packaged meats. This occurs because for many of these products, the price will consistently change during the 11-week rolling period to such an extent that one-third of the prices will not equal the modal price and therefore a new regular price is not calculated, and instead the previous regular price is carried forward. This often happens persistently for multiple years, so that a previous outdated regular price is carried on for a prolonged period, even though the price may have changed dozens of times over the same period. For example, when the algorithm was applied to store 122 from the Dominick’s dataset, [Midrigan \(2011\)](#) reported that 96.5% of price changes are temporary, 34.0% of prices change weekly, but only 2.9% of regular prices change weekly. Therefore, this approach, like that of [Eichenbaum et al. \(2011\)](#), yields aggregate price rigidity measurements that are unrealistically sticky.

The simple market reference price algorithm proposed in this paper provides a solution to overcome the previously outlined limitations associated with the aforementioned price-filtering algorithms. Market reference prices are defined as the modal price of all identical products sold in all retail stores across the United States during the same week. The algorithm computes a product’s market reference price by calculating the modal price of all units of a product sold that week across the 71,249 stores in the NielsenIQ retail scanner dataset. These calculations are possible because all units of a product contain the same UPC barcode regardless of what retail store it is sold in. A lower threshold of at least seven stores recording purchases of the UPC barcode during the week must occur for the market reference price to be calculated, otherwise the price observation was deleted from the dataset. This requirement is imposed to ensure that sufficient price information is present to calculate an informative market reference price, even for niche products.

The algorithm effectively mitigates any imperfections in the NielsenIQ retail scanner dataset. For example, any bias present in a product’s price recorded in a given store due to weekly average prices is removed after collapsing the data onto a nationwide weekly mode of the product’s price. [Cavallo \(2018\)](#) outlined how this bias changes the shape of the price-change distribution by turning larger price changes from the tails into smaller ones at the centre. Figure IV presents the distribution of posted price-change sizes and market reference price-change sizes in the NielsenIQ retail scanner dataset. The posted price distribution exhibits a roughly bell-shaped pattern centred near zero with moderate peakedness. In contrast, the market reference price distribution displays a more bimodal shape with heavier tails and some pronounced clustering at discrete intervals. These results illustrate how the market reference price algorithm can be used to correct the price-change size distribution in scanner data.

Figure IV: Distribution of Price Change Sizes



Notes: This figure presents the distribution of price-change sizes for posted prices (left panel) and market reference prices (right panel) across all observations in the NielsenIQ retail scanner dataset over 2006–2023.

When these price filtering algorithms were tested on the video and computer games product module, they produced a wide range of results. The unfiltered weekly posted price-change frequency was 46.09%, while the weekly market reference price-change frequency was 16.90%. The weekly regular price-change frequency result produced from the [Nakamura and Steinsson \(2008\)](#) sales filter A was 22.85%, while the weekly regular price-change frequency result produced from the [Kehoe and Midrigan \(2015\)](#) algorithm was 1.97%. The original version of the [Eichenbaum et al. \(2011\)](#) reference price algorithm will delete every single UPC in the product module due to its extreme requirement that UPCs must occur in at least 12 weeks of every calendar quarter. When these extreme restrictions are relaxed so that only one single calendar quarter must meet the requirement for 12 weekly occurrences, a very small fraction of UPCs are retained, but the weekly reference price-change frequency is only 0.16%, highlighting how this algorithm risks sample-selection bias.

In contrast, the market reference price algorithm provides a simple and effective mechanism to purge temporary price deviations from price rigidity measurements. This is because these temporary price deviations are idiosyncratic across retail chains. [Nakamura \(2008\)](#) found that for a given product, only 16% of its price variation is common across all stores nationwide, 65% is common to stores within the retail chain and 17% is idiosyncratic to individual stores. [DellaVigna and Gentzkow \(2019\)](#) found that within U.S. retail chains, there is nearly uniform pricing for identical retail products across stores, but also that there is substantial variation in the timing and depth of temporary sales across those same stores. A rationale for these findings is that retailers’ choices about the timing of temporary sales are strategic substitutes as they prefer to offer these price discounts when their competitors are not²⁶.

²⁶The intuition is that retailers sell to two types of consumers: price-insensitive buyers who purchase for immediate consumption, and bargain hunters who purchase the product to add to their stockpiles when prices are low. Retailers maximise their profits by offering temporary price discounts to attract bulk purchases from bargain hunters. However, if nearby competitors discount at the same time, this pricing strategy is undermined.

These findings explain why the algorithm, by generating nationwide central price measures which capture structural changes in a product’s price, performs well for products with both sticky and volatile price dynamics. Therefore, by examining a product’s price horizontally across all stores rather than focusing on a product’s price in a single store vertically over time, the algorithm enables the accurate calculation of aggregate price rigidity measurements for the United States. In Section V, we implement the market reference price algorithm across the entire NielsenIQ retail scanner dataset to measure price stickiness in the U.S. retail sector.

V Measuring Price Rigidity across Products and Time

The real economic effects of a monetary disturbance will not fully dissipate until the prices of all products adjust to reflect the changed monetary conditions. Therefore, the frequency at which prices change is a central issue in macroeconomic theory. This section seeks to answer the seemingly straightforward question: how often do the prices of products change? This task is unfortunately complicated by several factors including temporary price deviations, choice of price-filtering algorithm, product substitutions and the significant cross-sectional heterogeneity of price stickiness across product types. These challenges are evidenced by the significantly different price-change frequency results of [Nakamura and Steinsson \(2008\)](#) and [Klenow and Kryvtsov \(2008\)](#) who both analysed similar data from the CPI-RDB and used the BLS sales flag to calculate their regular price measurements²⁷. This is primarily due to the diverging weighting strategies employed in these studies. However, as [Nakamura and Steinsson \(2008, p. 1418\)](#) state: “There is no model-free way of selecting what is the appropriate summary statistic to describe the amount of aggregate price flexibility in an economy in which the frequency of price change varies across sectors from over 90% per month to less than 5% per month”. [Bils and Klenow \(2004\)](#) also found evidence for this tremendous cross-sectional heterogeneity of price stickiness where their monthly price-change frequency calculations ranged from 1.2% for coin-operated laundromats to 79% for unleaded gasoline.

Nonetheless, this study has sought to impose as few assumptions and constraints on the data as possible when computing aggregate price-change frequency measurements. The novel market reference price algorithm provides a simple and accurate mechanism to clean the price micro-data of temporary price deviations. Price-change frequencies were calculated using simple means within each of the included 1,100 product modules, and unweighted means and medians were then taken across these product modules to obtain product department and total aggregate estimates. This approach was carefully chosen to produce a central measure of price-change frequency that is not inadvertently skewed by any particular weighting strategy. Relatedly, [Kang and Usher](#)

²⁷See [Nakamura and Steinsson \(2013\)](#), for a detailed comparison of these results.

(2023) randomly sampled from 30 product modules in the NielsenIQ retail scanner dataset to show that products which generate more revenue tend to exhibit more frequent price-changes²⁸.

Furthermore, because each product has a unique UPC in the NielsenIQ retail scanner dataset, price rigidity measurements will not be biased by product substitutions²⁹. This stands in contrast to the unpublished BLS price micro-datasets underlying the CPI, where once a product's lifetime concludes, BLS pricing field agents are instructed to substitute it with the most similar available product in the store instead. Moreover, product categories such as apparel and transportation goods, which Nakamura and Steinsson (2008) found to experience high rates of substitution, are not included in the NielsenIQ retail scanner dataset.

Table II reports the mean and median weekly price-change frequency estimates for each of the ten product departments, as well as the aggregate total, in the NielsenIQ retail scanner dataset. Posted prices are the raw, unfiltered prices recorded when a product is sold at an electronic point-of-sale system in a retail store. Market reference prices are the mode of all posted prices recorded that week across all retail stores for an identical product. As documented by Nakamura and Steinsson (2013), the prevailing convention in macroeconomic literature has been to calibrate models to median rather than mean estimates of price-change frequencies. Moreover, unweighted median estimates provide a conceptually clear and robust measure of central tendency across the vast NielsenIQ retail scanner dataset, because they remain stable even in the presence of extreme outlier product modules³⁰. However, there is minimal difference between the mean and median estimates reported in Table II. The mean frequencies of posted and market reference price-changes are 29.70% and 17.12%, respectively, while the corresponding medians are 29.47% and 16.38%. Thus, filtering out temporary price deviations nearly halves the observed frequency of price-changes. This pattern is consistent with Nakamura and Steinsson (2008), who show that the estimated median price-change frequency of their regular price measure is roughly half that of their posted price estimates.

This median weekly market reference price-change frequency estimate of 16.38% indicates that retail prices in the United States are only mildly sticky. It is important to note, however, that other sectors of the economy appear to demonstrate differing dynamics. Bils and Klenow (2004) and Nakamura and Steinsson (2008) show that most service prices tend to be much stickier, whereas

²⁸In their preferred specification, they find that a 10% increase in revenue increases the probability of price adjustment by 0.13 percentage points.

²⁹Furthermore, the economics rationale is not as clear for excluding price-changes resulting from product substitutions as it is for temporary sales. For example, for product categories with high product turnovers, such as video games, the primary mechanism of price adjustment is when a new product is introduced. Manufacturer's suggested retail prices may remain unchanged until the release of the next title and only be altered to reflect current monetary and economic conditions at this time.

³⁰In particular, the candy-chocolate, soft-drinks-carbonated, candy-non-chocolate, soft-drinks-low-calorie, and cookies product modules in the Dry Grocery product department can impose significant distortive effects on the estimates if not properly accounted for. A detailed analysis of these effects is provided in the supplementary appendix.

transportation-related goods and services such as airline fares, new motor vehicles and fuel prices tend to be much more flexible. There is also a significant amount of heterogeneity in the frequency of posted and market reference price-changes across product modules in the NielsenIQ retail scanner dataset, which is illustrated in Figure V. Fresh strawberries is the product module with the most flexible posted prices (70.1%) and market reference prices (43.2%)³¹; while canned ham patties is the product module with the stickiest posted prices (4.1%) and magazines is the product module with the stickiest market reference prices (0.3%)³². This last finding is particularly noteworthy, given that [Cecchetti’s \(1986\)](#) study of magazine prices served as a predominant empirical reference for U.S. price-change frequencies until the publication of [Bils and Klenow \(2004\)](#)³³. This significant heterogeneity in price-change frequencies across retail products has implications for monetary non-neutrality. Demonstrably, in the multisector menu cost model of [Nakamura and Steinsson \(2010\)](#), the introduction of heterogeneity in the frequency of price-changes triples the degree of monetary non-neutrality generated by the model.

Table II: Price Change Frequency across Product Modules

Department	Posted Price		Market Ref.	
	Mean	Median	Mean	Median
Health and Beauty Aids	31.89	31.57	16.95	16.29
Dry Grocery	26.38	25.89	16.58	15.98
Frozen Foods	34.39	36.43	20.65	19.78
Dairy	36.66	36.98	20.19	19.67
Deli	33.70	35.48	15.07	12.76
Packaged Meat	37.52	45.26	21.37	22.69
Fresh Produce	34.17	37.58	19.65	18.74
Non-Food Grocery	27.18	28.85	14.95	14.49
Alcohol	26.10	25.87	20.75	19.69
General Merchandise	33.12	32.93	16.39	15.43
ALL	29.70	29.47	17.12	16.38

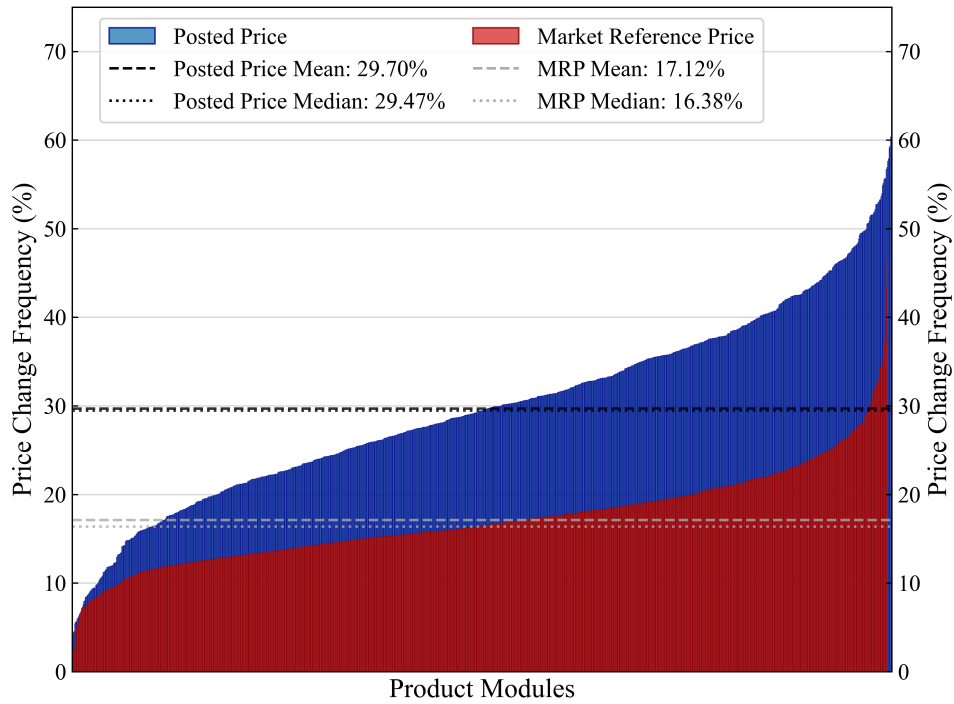
Notes: This table reports unweighted mean and median price change frequencies calculated across product modules within each product department in the NielsenIQ retail scanner dataset over the period 2006–2023. Price-change frequency statistics were calculated as the percentage of week-to-week observations where the price changed. Price-change frequency statistics were calculated using 1,100 of 1,116 total product modules; 16 product modules with sparse temporal coverage are excluded. A posted price refers to a product’s actual transaction price, while a market reference price is calculated as the mode of all the posted prices recorded for an identical product sold across all retail stores during the same week.

³¹Fresh strawberries have many price changes because they have a very short shelf life, highly seasonal growing seasons and complex supply chains which create volatile wholesale prices.

³²Magazine market reference prices are so sticky because publishers make most of their revenue and profits from the advertisements placed within the magazine. Therefore, it is important to maintain high retail sales volumes by fixing low retail prices over extended periods.

³³This finding that magazines have the stickiest market reference prices across all product modules, strongly suggests that the [Cecchetti \(1986\)](#) approach to estimate aggregate price-setting behaviour was flawed. Relatedly, [Blinder et al. \(1998\)](#) stated that the huge number of times that [Cecchetti’s \(1986\)](#) paper on magazine prices gets cited is ironic testimony to just how thin our knowledge is.

Figure V: Price Change Frequency across Product Modules



Notes: This figure presents two bar charts where each bar represents the weekly price-change frequency for an individual product module, for both posted prices (top panel) and market reference prices (bottom panel). The figure includes 1,100 of the 1,116 product modules in the NielsenIQ retail scanner dataset, as 16 tiny product modules were excluded due to extreme sparsity. A further 6 tiny product modules were excluded for market reference prices due to insufficient store coverage.

A common misconception is that monthly price-change frequency estimates derived from CPI price micro-datasets, such as the BLS CPI-RDB, capture all the price-changes which occur in a month. However, in practice, BLS field agents collect the price of a product or service only once in the month, so intra-month changes are unobserved. Therefore, studies which rely on such data derive their price rigidity estimates by comparing weekly price-change frequencies at a monthly lag. [Cavallo \(2018\)](#) showed that such monthly sampling makes prices appear stickier because many price-changes within the month are not observed. Furthermore, temporary price reductions tend to have durations shorter than four weeks³⁴.

A more accurate measure of the repricing of the economy in response to nominal shocks is to calculate the share of products that undergo at least one weekly price-change within a month. This measure captures the share of products that have at least one opportunity to reprice in the month following a nominal shock. Table III reports the share of products within the NielsenIQ retail scanner dataset which have at least one weekly price-change within a month. The calculation methodology employed mirrors the approach adopted in Table II. The mean frequencies of

³⁴[Seaton and Waterson \(2013\)](#) provide an example from scanner data for British supermarkets which documents this feature of temporary price reductions.

posted and market reference price-changes are 55.32% and 41.61%, respectively, while the corresponding medians are 56.00% and 41.55%. These results show that the moderately high weekly price-change frequency results reported in Table II can give a misleading impression of overall price flexibility. Price changes are not independent over time; a subset of products undergo repeated weekly price adjustments, while many others remain unchanged for prolonged periods. This uneven distribution of adjustments induces overall price stickiness and helps explain why nominal shocks can produce persistent real economic effects.

Table III: Share of Products with ≥ 1 Weekly Price Change per Month

Department	Posted Price		Market Ref.	
	Mean	Median	Mean	Median
Health and Beauty Aids	48.72	50.42	41.37	41.22
Dry Grocery	54.76	54.85	39.30	39.15
Frozen Foods	64.79	64.90	50.23	50.12
Dairy	63.23	62.96	46.28	46.05
Deli	62.58	62.56	35.47	35.48
Packaged Meat	66.98	67.66	48.06	47.98
Fresh Produce	52.83	53.65	36.97	37.01
Non-Food Grocery	50.54	51.24	38.45	38.40
Alcohol	48.75	48.57	45.51	45.22
General Merchandise	43.17	45.93	36.01	36.13
All Departments	55.32	56.00	41.61	41.55

Notes: This table reports the percentage of products with at least one weekly price change per month in the NielsenIQ retail scanner dataset. All statistics are calculated from February 2006 through December 2023. January 2006 is excluded due to downward bias present in the first month. All means and medians are unweighted calculations across the product modules within each product department. Price-change frequency statistics were calculated using 1,100 of 1,116 total product modules; 16 product modules with sparse temporal coverage are excluded. A posted price refers to a product's actual transaction price, while a market reference price is calculated as the mode of all the posted prices recorded for an identical product sold across all retail stores during the same week.

Figure VI presents the development of price-change frequencies for both posted and market reference prices between 2006–2023. There is a strong upwards trend in the frequency of posted price changes. The posted price change frequency increased from 22.28% to 39.46% in the last week of the years 2006 and 2023 respectively. In contrast, there is no evident upwards trend present in the frequency of market reference price changes. The key difference between the two time series is that market reference prices, by their calculation, are purged of temporary price deviations, whereas posted prices are not. This suggests that the observed steady decline in price stickiness is largely due to an increase in the prevalence of temporary price deviations. Two potential contributory factors to this interesting upwards trend are the simultaneous rise in online shopping and the use of computer algorithms for automatic price-setting. According to the United States Census Bureau's Quarterly E-Commerce reports, online shopping as a percentage of total

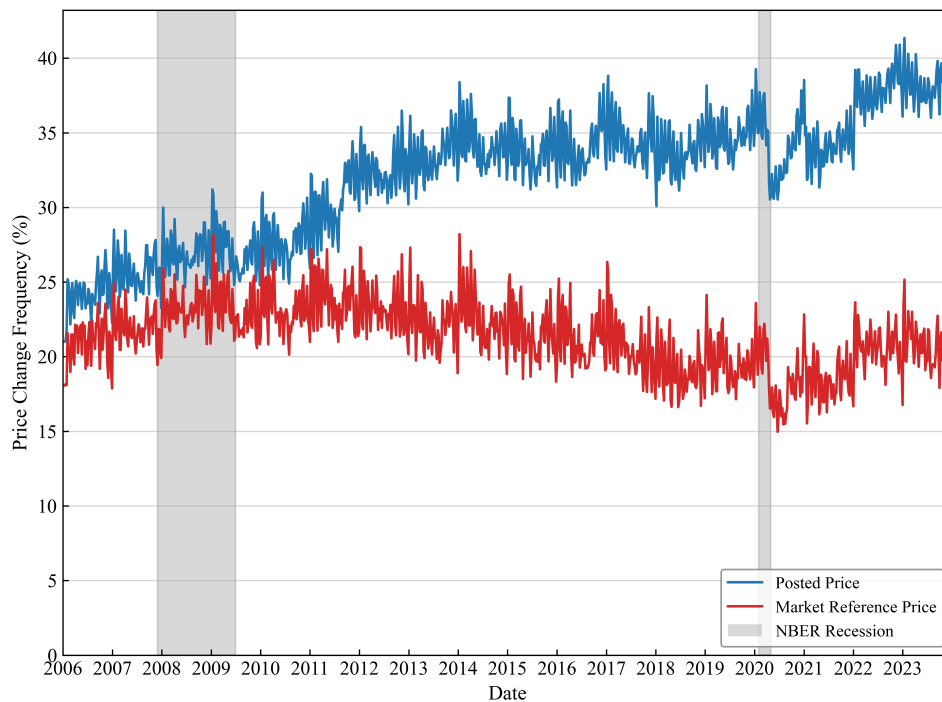
retail sales increased from 2.9% in 2006 to 15.3% in 2023³⁵. These technological advancements have enhanced retailers' capability to implement short-run promotions and dynamic pricing while also substantially lowering the managerial and menu costs of price adjustments³⁶. The increase in the prevalence of temporary price deviations may be part of an even longer-run trend that predates the beginning of the sample period. Consistent with this interpretation, [Nakamura et al. \(2018\)](#) document a significant increase in the frequency of temporary sales between 1978–2014 in the processed and unprocessed food, apparel, household furnishings, and recreation goods sectors of the CPI-RDB.

The consensus in macroeconomic theory has been that temporary sales have minimal implications for monetary non-neutrality. However, this consensus has recently been questioned by [Sudo et al. \(2018\)](#) and [Kryvtsov and Vincent \(2021\)](#), who provide empirical evidence that retailers' decisions to conduct temporary sales are not orthogonal to economic conditions. Using a large retail scanner dataset, [Sudo et al. \(2018\)](#) document a strong upwards trend in temporary sales in Japan between 1988–2013, which they attribute to deteriorating labour market conditions during the “Lost Decades”. [Kryvtsov and Vincent \(2021\)](#) similarly find strong co-movement between the incidence of temporary sales and unemployment using U.K. and U.S. CPI microdata. Therefore, our finding that technological advancement is likely increasing the prevalence of temporary sales in the United States, suggests that their macroeconomic impact is becoming increasingly pertinent. Therefore, study of their impact on monetary non-neutrality should remain a key area of focus for future theoretical macroeconomic research.

³⁵Furthermore, [Cavallo \(2017\)](#) found that 69% of products' online prices match their offline price in the United States, indicating that retail sales data should reflect broadly similar pricing patterns.

³⁶[Zbaracki et al. \(2004\)](#) identify that in addition to menu costs, firms also have three types of managerial costs (information gathering, decision-making and communications costs) and two types of customer costs (communication and negotiation costs) which impede price adjustments.

Figure VI: Price Change Frequency across Time



Notes: This figure presents a line chart of the mean weekly posted price- and market reference price-change frequency statistics over 2006–2023. Mean price-change frequency statistics were calculated for every one of the individual 1,100 product modules each week, the aggregate statistic was then calculated as the unweighted mean across all 1,100 product modules. 16 product modules with sparse temporal coverage are excluded. A posted price refers to a product’s actual transaction price, while a market reference price is calculated as the mode of all the posted prices recorded for an identical product sold across all retail stores during the same week. The grey shading represents time-periods where the NBER recorded that the U.S. economy was in recession.

Another important point evident from Figure VI is that there does not appear to be any discernible cyclicity in the price rigidity time series. Notably, there was no apparent response in price-change frequencies during the Global Financial Crisis or the subsequent Great Recession. This finding undermines predictions regularly made in macroeconomic theory, that recessions are longer lasting than economic expansions in part because price rigidity intensifies countercyclically during downturns. It also undermines econometric models that are designed to identify the response of aggregate price rigidity to exogenous structural shocks, as time series which exhibit no cyclical variation cannot yield meaningful impulse responses. However, there is a clear downward shift in price-change frequencies during the Covid-19 pandemic, aligning with both the nadir of grocery and drugstore footfall and the height of government price-gouging enforcement efforts³⁷. Price-change frequencies rise during the 2021–2023 high inflation episode, but this increase appears largely consistent with the continuation of pre-pandemic trends. Nevertheless,

³⁷This sudden drop in price-change frequencies occurs during the week ending on Saturday 18th April 2020. See Ritchie (2020) for a detailed analysis of the Google mobility trends of retail store footfall during this period.

the timing suggests that elevated inflation exerted some additional upward influence on the frequency of price adjustment. Section VI shows that this effect operates primarily through an increase in the frequency of price increases.

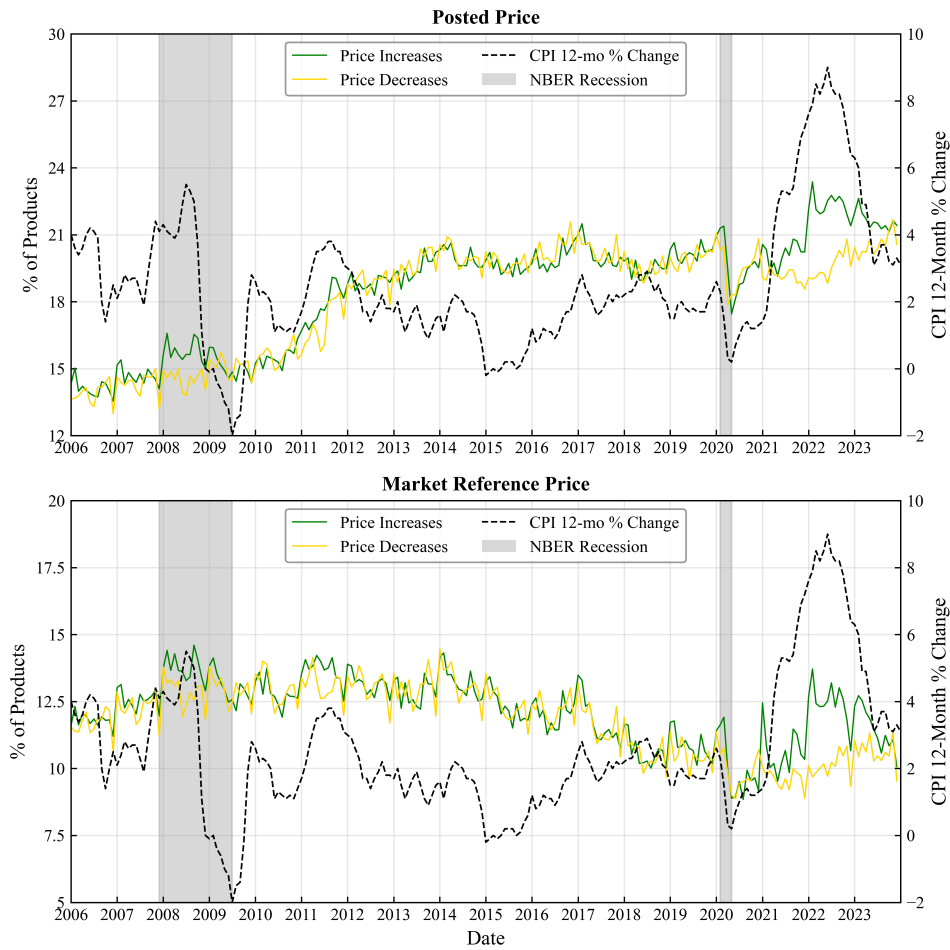
VI The Dynamics of Price Changes

Inflation reflects the aggregation of firm-level price-setting behaviour across the economy. Aggregate price change dynamics can be decomposed into four mechanical levers: the frequency of price increases, the frequency of price decreases, the average size of price increases, and the average size of price decreases. In a high-inflation environment, state-dependent sticky price models, such as menu cost models, imply that inflation is driven primarily by more frequent price increases and less frequent price decreases, whereas time-dependent sticky price models imply that it is driven primarily by larger price increases and smaller price decreases. This section examines the relative importance of these four mechanical levers in generating inflation in order to assess the empirical validity of alternative theories of price stickiness.

Figure VII presents the development of the frequency of price increases and decreases for posted prices (top panel) and market reference prices (bottom panel) between 2006–2023, alongside the CPI inflation rate. Inflation is measured using the seasonally adjusted 12-month percentage change in the consumer price index for all Urban Consumers (CPI-U), which was obtained from the BLS. During the low-inflation period between 2009–2020, movements in the frequency of price increases and decreases largely offset one another, resulting in limited net inflationary pressure. In contrast, during the high-inflation episode between 2021–2023, there is a pronounced and sustained increase in the frequency of price increases relative to price decreases, coinciding with the elevated inflation. There is also a discernible rise in the frequency of price increases above the frequency of price decreases during the early phase of the Global Financial Crisis, when inflation briefly surged above 5%. These patterns suggest that the frequency of price increases becomes an increasingly important determinant of inflation when inflation reaches elevated levels. They also help explain the limited apparent cyclicity in the aggregate frequency of price-changes observed during the high-inflation episode presented in Figure VI.

Figure VIII presents the development of the mean size of price increases and decreases for posted prices (top panel) and market reference prices (bottom panel) between 2006–2023, alongside the CPI inflation rate. Inflation is measured using the seasonally adjusted 12-month percentage change in the CPI-U, which was obtained from the BLS. Throughout the full sample period, the mean size of price increases consistently exceeds that of price decreases. This finding indicates that during low-inflation periods, the average size of price increases serves as the predominant determinant of inflation. However, during the high-inflation episode between 2021–2023, there is no evidence of a widening in the difference between the average sizes of price increases and

Figure VII: Price Change Direction Time Series

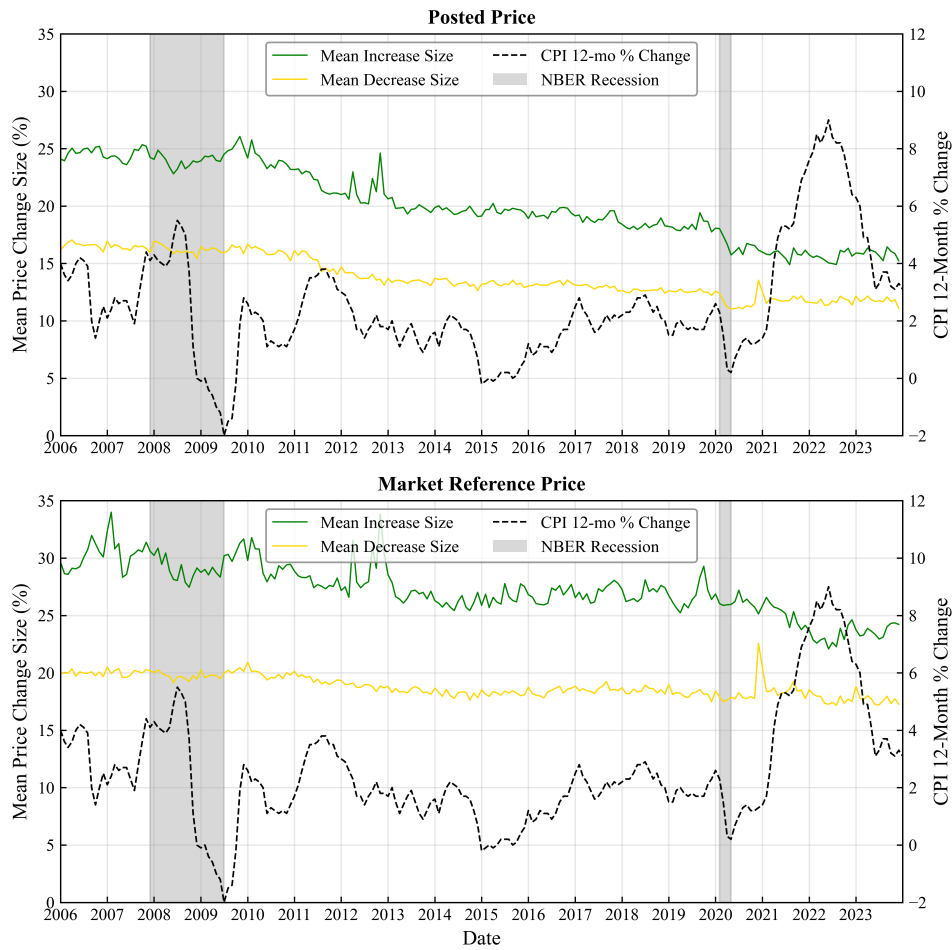


Notes: This figure presents the monthly percentage of observations in the NielsenIQ retail scanner dataset that are price increases (green line) and price decreases (gold line), over 2006–2023, for posted prices (top panel) and market reference prices (bottom panel), with an overlay of the BLS CPI-U 12-month percentage change (black dashed line). The grey shading represents time-periods where the NBER recorded that the U.S. economy was in recession.

decreases. These patterns suggest that during high-inflation periods, the average size of price increases continues to contribute to inflation, but becomes a relatively less important determinant than the frequency of price increases as inflation rises.

Overall, these findings are broadly consistent with those found in the previous academic literature. [Nakamura et al. \(2018\)](#) documented that the frequency of price increases surged in the United States during the Great Inflation of the 1970s and 1980s. They also found that the frequency of price-changes fell during the subsequent Great Moderation, driven entirely by a fall in the frequency of price increases. [Wulfsberg \(2016\)](#) documented similar trends in Norway over the same two periods. They found that when inflation was high and volatile, prices changed more frequently and in small magnitudes, whereas when inflation was low and stable, prices changed less frequently but in larger magnitudes. [Gagnon \(2009\)](#) studied price-setting behaviour

Figure VIII: Price Change Size Time Series



Notes: This figure presents the monthly mean absolute size of price increases (green line) and price decreases (gold line) in the NielsenIQ retail scanner dataset over 2006–2023, for posted prices (top panel) and market reference prices (bottom panel), with an overlay of the BLS CPI-U 12-month percentage change (black dashed line). The grey shading represents time-periods where the NBER recorded that the U.S. economy was in recession.

in Mexico over 1994–2002, a period characterised by a sharp inflation surge spurred by the peso crisis followed by a gradual return to low and stable inflation. They documented that when inflation was below 10–15%, the frequency of price changes co-moved weakly with inflation, as increases and decreases largely offset one another, particularly for goods. In this range, inflation was instead closely related to the average magnitude of price changes. In contrast, when inflation rose above 10–15%, few price decreases were observed and both the frequency of price increases and their average magnitude emerged as important determinants of inflation.

Table IV reports the mean weekly price increase and decrease frequency estimates for each of the ten product departments, as well as the aggregate total in the retail scanner dataset. Table V reports the corresponding mean sizes of weekly price increases and decreases for each product department and for the aggregate. Over the full sample period from 2006–2023, the average size of price increases, rather than the frequency of price increases, acts as the predominant

determinant of inflation. This reflects the fact that the sample period is dominated by episodes of low and stable inflation. Similar conclusions are reached in previous studies which focus primarily on low-inflation sample periods, including [Klenow and Kryvtsov \(2008\)](#) which studied U.S. price-setting behaviour between 1988–2004 and [Gautier et al. \(2024\)](#) which analysed euro area price-setting behaviour between 2010–2019. The sensitivity of these results to the chosen sample period highlights the importance of examining price-setting behaviour across both low- and high-inflation periods. Consistent with the evidence presented in Figures VII and VIII, inflation is primarily driven by the size of price increases in low-inflation periods, whereas at elevated inflation rates it is increasingly driven by the frequency of price increases.

Table IV: Price Change Direction Frequency across Product Departments

Department	Posted Price		Market Reference Price	
	Increase (%)	Decrease (%)	Increase (%)	Decrease (%)
Health and Beauty Aids	18.00	17.65	12.16	11.68
Dry Grocery	18.52	18.13	12.08	11.71
Frozen Foods	21.62	21.33	15.71	15.40
Dairy	20.89	20.41	13.13	12.82
Deli	21.34	21.03	12.93	12.68
Packaged Meat	22.62	21.85	14.86	14.52
Fresh Produce	19.26	18.45	10.49	10.11
Non-Food Grocery	18.01	17.33	11.62	11.07
Alcohol	16.26	15.77	11.26	10.92
General Merchandise	16.07	16.53	8.66	8.89
Total	18.57	18.20	12.17	11.81

Notes: This table reports the percentage of weekly price observations which were increases or decreases in the NielsenIQ retail scanner dataset over the period 2006–2023. A posted price refers to a product’s actual transaction price, while a market reference price is calculated as the mode of all the posted prices recorded for an identical product sold across all retail stores during the same week.

Overall, the patterns of price-setting behaviour documented in this section have important implications for the microeconomic foundations of price rigidity underlying New Keynesian macroeconomic theory. The [Calvo \(1983\)](#) price-setting framework assumes that the frequency of price changes is fixed over time, which implies that the average size of price increases will rise sharply as inflation surges. This model can adequately approximate price-setting behaviour during low-inflation periods, where inflation is primarily driven by the average size of price increases rather than the frequency of price increases. However, in high-inflation periods such as the high-inflation episode of 2021–2023, or the Great Inflation, these assumptions become less tenable. During these periods, inflation is increasingly driven by the frequency of price increases, which contradicts the [Calvo \(1983\)](#) assumption that a fixed fraction of firms adjust their prices each period independent of economic conditions. Moreover, there is no evidence of a large expansion in the average size of price increases during these high-inflation periods, which the [Calvo \(1983\)](#) model would imply.

Table V: Average Price Change Sizes by Department

Department	Posted Price		Market Reference Price	
	Increase (%)	Decrease (%)	Increase (%)	Decrease (%)
Health and Beauty Aids	23.94	-16.38	30.18	-20.79
Dry Grocery	21.28	-14.45	28.01	-19.50
Frozen Foods	21.50	-14.83	26.66	-19.14
Dairy	16.95	-11.87	23.71	-17.32
Deli	21.22	-14.55	25.50	-18.43
Packaged Meat	21.60	-15.03	26.26	-18.92
Fresh Produce	20.51	-13.54	30.85	-20.63
Non-Food Grocery	16.43	-11.58	24.44	-17.51
Alcohol	11.40	-9.37	14.32	-11.86
General Merchandise	26.83	-16.31	41.51	-23.00
Total	20.17	-13.83	27.28	-18.81

Notes: This table reports the average percentage size of price increases and decreases in the NielsenIQ retail scanner dataset over the period 2006–2023. A posted price refers to a product’s actual transaction price, while a market reference price is calculated as the mode of all the posted prices recorded for an identical product sold across all retail stores during the same week.

In contrast, menu cost models, which assume that all firms can adjust their prices each period subject to the payment of adjustment costs, can effectively approximate price-setting behaviour during both low- and high-inflation periods. When inflation is low, most firms will tolerate small deviations from their optimal price. As a result, the fraction of firms adjusting their prices remains largely stable and inflation primarily reflects the average size of price-changes among the firms that do adjust. When inflation is high, more firms will increase their prices because the benefit derived from adjusting to the new optimal price will exceed the adjustment costs. Consequently, as inflation rises, it becomes increasingly driven by the frequency of price increases rather than their average size.

This finding that menu cost models better reflect price-setting behaviour in the United States, particularly during high-inflation periods, has important implications for optimal monetary policy. [Burstein and Hellwig \(2008\)](#) established the result that the welfare costs of inflation are much higher in an economy with Calvo-style price staggering, than in a menu cost economy. The key mechanism underlying this result is the [Calvo \(1983\)](#) assumption that a fixed fraction of firms receive an opportunity to adjust their prices each period, independent of economic conditions. As inflation rises, this implies that some firms’ prices will remain far from their optimal levels for extended periods of time. In contrast, menu cost models allow all firms to adjust their prices each period subject to adjustment costs, limiting the extent to which prices will drift away from their optimal level. As a result, monetary policy rules derived under the [Calvo \(1983\)](#) price-setting framework may overstate the welfare gains from aggressive price stabilisation during high-inflation episodes.

VII Conclusion

In this paper, we analyse retailer price-change dynamics across the full U.S. business cycle. This is achieved by applying a novel market reference price algorithm across the immense NielsenIQ retail scanner dataset. By examining a product’s price horizontally across all stores rather than focusing on a product’s price in a single store vertically over time, the algorithm enables the accurate calculation of aggregate price rigidity measurements for the United States. Altogether, we analyse over 239 billion prices from over 1,100 categories of retail products which were collected from 71,249 retail stores between 2006–2023. We find that: (1) aggregate market reference prices are only mildly sticky, but there is substantial heterogeneity in price stickiness across retail products; (2) while posted price-change frequency exhibits a strong upward trend over 2006–2023, market reference price-change frequency remains largely stable; and (3) in low-inflation periods, variations in inflation are primarily accounted for by changes in the size of price adjustments rather than their frequency, whereas in high-inflation periods, the frequency of price increases becomes the dominant margin of adjustment. This last finding in particular highlights the need to construct more robust microeconomic foundations of price rigidity for New Keynesian macroeconomic theory that better reflect actual U.S. price-setting behaviour throughout the business cycle.

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